MSF-MOFS: Budget-Efficient Multi-Scale Fisher Seeding for High-Dimensional Multi-Objective Feature Selection

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Abstract. High-dimensional wrapper-based feature selection under tight evaluation budgets faces a recurring trade-off: prior guidance can accelerate search, yet richer guidance increases preprocessing cost and depends on relevance estimates that may be unreliable. This problem also arises in satellite telemetry. Multi-Scale Fisher Seeding for Multi-Objective Feature Selection (MSF-MOFS) treats prior knowledge as a bounded information budget: one Fisher ranking on the complete outer-training split constructs exactly three guided chromosomes at complementary absolute-relative sparsity scales, after which standard NSGA-II or SPEA2 proceeds unchanged. Across nine public datasets spanning 100 to 22,283 features, MSF achieves higher mean held-out Pareto-set hypervolume than matched Random under both carriers, with 87/90 paired wins and eight of nine Holm-significant datasets per carrier, while remaining competitive at lower preprocessing cost. Sparse-control shows that multi-scale placement is robust, while feature identity depends on ranking reliability. In a controlled satellite application, 12 channels over a 64-point causal history yield 582 candidate features; MSF outperforms matched Random in all 40 paired subsystem runs, and a frozen 140-feature plan reduces median online extraction time by 59.8% while retaining useful fault evidence. These results support fixed-count, multi-scale, initialization-only guidance as a reusable population-level interface for budget-constrained evolutionary feature selection.

Keywords: Multi-objective feature selection, Evolutionary initialization, Fisher score, High-dimensional classification, Satellite telemetry

1 Introduction

High-dimensional wrapper-based feature selection (FS) is constrained by two scarce resources: model-evaluation budget and reliable prior information. Evolutionary FS can optimize predictive quality and subset size directly, but each candidate subset requires model training [4,19]. Random starts can therefore waste evaluations, whereas richer guidance—filter ensembles, repeated relevance estimation, or

problem-specific operators—adds preprocessing and implementation cost [2,13,15,18,20]. The central problem is to obtain a substantial search advantage from prior information without making the guidance mechanism expensive or persistent.

Satellite telemetry provides a engineering instance. Temporal and cross-channel representations are increasingly used because faults may appear as drifts, transients, or changes in channel relationships [1,10,16]. In our controlled case, 12 channels over a 64-point causal history produce 582 candidate features. The resulting search space is much larger than the raw space, and different subsystem faults rely on different feature relationships, making compact upstream feature selection operationally relevant.

MSF-MOFS introduces a bounded population-design interface: a single Fisher ranking places exactly three guided chromosomes at complementary absolute-relative sparsity scales. The guidance count is fixed and intervention ends at generation 0 so standard NSGA-II or SPEA2 receives size-diverse, relevance-informed starting solutions without changes to algorithm mechanism. This fixed-count, mixed-scale construction requires only one ranking and leaves the subsequent search unchanged.

The public datasets provide the main methodological evidence: effectiveness against stronger guided baselines, robustness from 100 to 22,283 features, reuse across two MOEA carriers, preprocessing efficiency, and mechanism attribution through Sparse-control and ARCENE. Together, these experiments support three contributions: a low-intervention initialization principle, evidence separating robust scale placement from reliability-dependent feature identity, and a cross-domain application without redesigning the downstream optimizer.

2 Related Work

2.1 Evolutionary Feature Selection under Expensive Evaluations

Evolutionary FS commonly encodes subsets as binary chromosomes and evaluates them with predictive models [19]. Wrapper formulations directly optimize downstream performance but become expensive as dimensionality grows. Many methods address high-dimensional or imbalanced settings [5,9,11,12,17]. Our focus is complementary: how to introduce useful problem information before search while keeping its cost and coupling low.

2.2 Search Guidance in High-Dimensional Evolutionary Feature Selection

Guided methods often couple initialization with additional search machinery. Segmented initialization is paired with offspring modification [18], HFMOEA uses ten filter rankings [13], E-NSGA-II adds specialized operators [15], and MBLPE combines Fisher analysis with predictive evolution [20]. By contrast, MSF seeds exactly three size-diverse, Fisher-informed solutions from a single ranking and leaves standard evolutionary operators unchanged after initialization.

2.3 Satellite Telemetry as a Structured Feature-Selection Setting

Spacecraft telemetry analysis increasingly models temporal and multivariate structure through forecasting, correlation-aware anomaly measures, and probabilistic health models [1,10,16]. Such representations improve fault sensitivity but also expand the upstream feature space. We use satellite telemetry specifically as a structured feature-selection setting: the question is whether a compact derived feature subset can be found efficiently under a constrained wrapper budget.

3 MSF-MOFS: Design Principle and Method

MSF treats prior knowledge as a bounded information budget. Three design choices are separated explicitly: guidance amount is fixed to three chromosomes, their cardinalities span complementary absolute-relative scales, and guidance acts only through the initial population (Fig. 1).

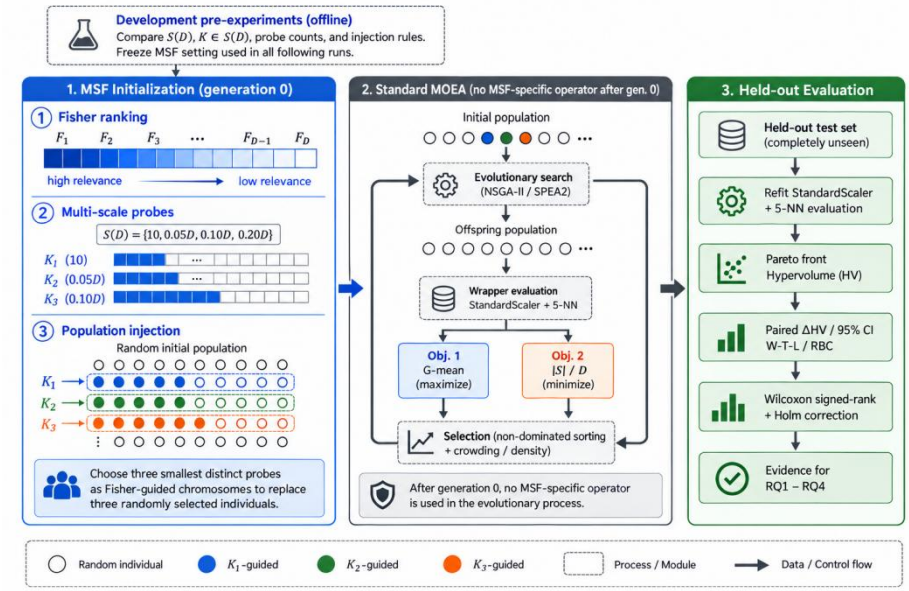


Fig. 1. Overall MSF-MOFS workflow.

3.1 Fixed-Count Information Injection

Exactly three informed chromosomes replace three individuals in a matched random population, after which control returns to the downstream MOEA. The same intervention occupies 25% of a population of 12 but only 6% of a population of 50, so information count—not guided fraction—is fixed. Within a dataset, one relevance

ranking is reused across all sparsity scales and both carriers; MSF changes only the starting population, not wrapper objectives, variation, or survivor selection.

3.2 Single-Ranking Relevance Prior

For binary labels $y \in \{p, n\}$, feature j receives a Fisher relevance score computed once on the complete outer-training split,

$$F_j = \frac{(\mu_{jp} - \mu_{jn})^2}{(\sigma_{jp})^2 + (\sigma_{jn})^2 + \varepsilon}. \quad (1)$$

where p and n denote the two classes; $\varepsilon = 10^{-12}$ is a fixed denominator stabilizer, and the means and variances are class-conditional statistics from the complete outer-training split. The implementation rejects non-binary labels. Sorting scores in descending order gives the dataset-specific ranking results, used only to determine feature identities in the three guided chromosomes.

3.3 Absolute-Relative Multi-Scale Construction

From a matched Bernoulli(0.20) population, MSF replaces exactly three chromosomes. Candidate cardinalities are drawn from

$$S(D) = \{10, 0.05D, 0.10D, 0.20D\} \quad (2)$$

Let the raw candidate multiset be $S_0(D) = \{10, 0.05D, 0.10D, 0.20D\}$. Each relative term is first rounded to an integer; every resulting value is then clipped to $[1, D]$; duplicates are removed; and the three smallest distinct valid cardinalities are retained, in that order, as $K(D)$. Exactly three chromosomes are replaced by the nested top-K Fisher prefixes, and no cardinality or ranking is recomputed after generation 0. This yields 5/10/20 for Hill-Valley, 8/10/17 for Musk1, 10/25/50 for MADELON, and 10/500/1000 for ARCENE. Sparse-control keeps $K(D)$ but randomizes feature identity.

3.4 Carrier-Independent Population Interface

After initialization, standard NSGA-II [4] runs unchanged. Cross-carrier evaluation passes the same binary population to standard SPEA2 [21] with matched variation, base population, and random seed. No MSF-specific operator is used after generation 0, so carrier validation tests the population prior rather than a carrier-specific mechanism.

4 Experimental Design

The evaluation addresses effectiveness (RQ1), dimensional robustness (RQ2), carrier reuse (RQ3), and guidance cost/occupancy (RQ4), followed by mechanism analysis and the satellite application.

4.1 Wrapper Stress Protocol and Objectives

For each outer-training split, training-only environments $R = \{1,3,5\}$ are built from a fixed reference-majority pool and nested minority pools, each with stratified three-fold partitions. Within every fold, StandardScaler is fitted on the training portion and followed by 5-NN. Validation uses the geometric mean of true-positive rate (TPR) and true-negative rate (TNR),

$$G_r(\mathbf{s}) = \sqrt{\mathrm{TPR}_r(\mathbf{s})\mathrm{TNR}_r(\mathbf{s})}. \tag{3}$$

All methods share the same bi-objective formulation,

$$f_1(\mathbf{s}) = 1 - \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} G_r(\mathbf{s}), \qquad f_2(\mathbf{s}) = \frac{\|\mathbf{s}\|_0}{D}. \tag{4}$$

Training-selected subsets are refitted and evaluated on the untouched outer test set. The held-out minimization objectives are $f_1 = 1 - G_{mean}$ and $f_2 = \frac{\|\boldsymbol{s}\|_0}{D}$; both are intrinsically bounded in [0,1], so no dataset- or carrier-specific post-hoc normalization is applied. The hypervolume (HV) reference point is fixed at (1,1) for every dataset and both carriers [22].

4.2 Datasets and Evidence Roles

Table 1 summarizes nine public high-dimensional binary datasets. MADELON [7], Hill-Valley [6], Musk1 [3], and ARCENE [8] form the core suite. Five additional datasets—ALLAML, colon, GLI-85, Prostate-GE, and SMK-CAN—were selected before MSF results were inspected using prespecified breadth criteria: binary classification, D≥2000, and 50≤N≤200 [14].

Table 1. Final nine-dataset evaluation suite and evidence roles.

Dataset	N	D	Class counts	Role
MADELON	2600	500	1300/1300	Core
Hill-Valley	1212	100	600/612	Core
Musk1	476	166	269/207	Core
ARCENE	200	10000	112/88	Core
ALLAML	72	7129	47/25	Breadth
colon	62	2000	40/22	Breadth
GLI-85	85	22283	26/59	Breadth
Prostate-GE	102	5966	50/52	Breadth

Dataset	N	D	Class counts	Role
SMK-CAN	187	19993	90/97	Breadth

4.3 Public-Dataset Protocol and Baselines

The mixed-scale rule was selected in a six-dataset development study (the core suite, Internet Advertisements, and SECOM). Seeds 82–91 compared mixed-scale schedules with m=1–4 and relative-only m=3. Relative-only m=3 led in numerical average rank but was significantly worse on Internet Advertisements, while mixed-scale m=3 remained competitive across datasets. The mixed-scale schedule was fixed for confirmation with ten new matched seeds; the five breadth datasets were evaluated only afterward. Unless stated otherwise: stratified 70/30 outer split, population 12, six generations, and fold-local StandardScaler+5-NN.

HFMOEA-CP retains ten-filter initialization [13] and MBLPE-CP is ported from the public implementation [20]; both use the same splits, wrapper, objectives, and search budget. Timing excludes loading, splitting, and shared environment construction.

4.4 Evaluation and Statistical Analysis

The primary outcome is held-out HV. Matched two-sided Wilcoxon tests use Holm correction within prespecified families: four core datasets per external baseline, nine datasets per carrier and carrier-interaction family, and five breadth datasets per Sparse-control contrast. Complete per-dataset MSF-versus-Random reporting includes wins/ties/losses (W/T/L), rank-biserial correlation (RBC), and paired 95% bootstrap CIs for mean ΔHV from 20,000 resamples of matched seed differences. For the satellite study, the same paired/Holm procedure is applied to four prespecified subsystem spaces.

4.5 Structured Satellite-Telemetry Application Protocol

The satellite study uses controlled synthetic telemetry. A causal 64-point history over 12 channels yields 582 candidates (288 univariate and 294 cross-channel). Candidate spaces are Power 267, Thermal 135, Attitude 135, Data 204, Communication 69, and Propulsion 24; the first four form the comparative MSF family, whereas Communication and Propulsion retain predesignated direct-feature treatment.

Feature-selection labels come from fault-injection ground truth: nominal windows are negative, established target-subsystem faults are positive, and transition windows are excluded. Each subsystem uses the same classification-quality versus feature-ratio wrapper; selected features are only inputs to a fixed downstream application, not direct health-score predictions.

Feature definitions, subsystem mappings, feature sets, and downstream settings were fixed before independent evaluation. The confirmation bank contains 411 scenario-disjoint cases and 9,042 windows. The primary comparison is MSF versus

matched Random under identical wrapper, objectives, and search budget, evaluated by held-out HV.

5 Results and Discussion

Results are organized around effectiveness, reuse, efficiency, mechanism, and structured application transfer.

5.1 RQ1: Effectiveness under Tight Wrapper Budgets

Table 2 compares MSF with more elaborate guided baselines. MSF ranks first on all four core datasets by mean HV. Against HFMOEA-CP, improvements are significant on Hill-Valley and ARCENE and non-significant on MADELON and Musk1; against MBLPE-CP, MADELON and ARCENE are significant. Average rank is 1.00 for MSF, 2.25 for HFMOEA-CP, and 2.75 for MBLPE-CP, supporting competitive quality without claiming universal pairwise dominance.

Table 2. Held-out Pareto-set HV (dimensionless; reference (1,1); higher is better), mean ± std.

Dataset	MSF-MOFS	HFMOEA-CP	MBLPE-CP
MADELON	0.7555±0.0098	0.7554±0.0098	0.6297±0.0441
Hill-Valley	0.3616±0.0188	0.3205±0.0213	0.3464±0.0229
Musk1	0.7075±0.0237	0.6996±0.0269	0.6902±0.0285
ARCENE	0.6728±0.0499	0.6057±0.0603	0.5665±0.0567
Avg. rank	1.00	2.25	2.75

5.2 RQ2-RQ3: Dimension-Spanning Robustness and Cross-Carrier Reuse

Across D=100 to 22,283, all nine dataset-level mean effects favor MSF under both carriers (Table 3), with mean dataset-level ΔHV≈+0.2743; 8/9 are Holm-significant and Musk1 is the only non-significant case. No carrier-interaction test is Holm-significant. Four outward HV-reference perturbations—(1.05,1.05), (1.10,1.10), (1.10,1.05), and (1.05,1.10)—each retain 9/9 positive mean gains and 8/9 Holm-significant datasets under both carriers.

Table 3. Held-out MSF−Random ΔHV (dimensionless; ref=(1,1); positive favors MSF). Cell: mean ΔHV [95% CI]; W/T/L; RBC; Holm p.

Dataset	NSGA-II	SPEA2
MADELON	+.408 [.391,.427];10/0/0;1.00;.0176	+.406 [.391,.423];10/0/0;1.00;.0176
Hill-Valley	+.026 [.012,.040];9/0/1;.855;.0273	+.035 [.023,.049];10/0/0;1.00;.0176
Musk1	+.029 [-.0001,.0586];8/0/2;.636;.0840	+.027 [-.0006,.0512];7/0/3;.636;.0840
ARCENE	+.163 [.116,.211];10/0/0;1.00;.0176	+.122 [.078,.173];10/0/0;1.00;.0176
ALLAML	+.504 [.433,.570];10/0/0;1.00;.0176	+.522 [.461,.568];10/0/0;1.00;.0176
colon	+.388 [.312,.461];10/0/0;1.00;.0176	+.416 [.332,.502];10/0/0;1.00;.0176

Dataset	NSGA-II	SPEA2
GLI-85	+.396 [.325,.476];10/0/0;1.00;.0176	+.381 [.308,.451];10/0/0;1.00;.0176
Prostate-GE	+.394 [.355,.428];10/0/0;1.00;.0176	+.397 [.358,.435];10/0/0;1.00;.0176
SMK-CAN	+.162 [.127,.197];10/0/0;1.00;.0176	+.164 [.130,.199];10/0/0;1.00;.0176

5.3 RQ4: Guidance Efficiency and Low-Occupancy Robustness

MSF does not move complexity into preprocessing. On MADELON, preprocessing is about 0.007 s for MSF versus 2.088 s for HFMOEA-CP (total runtime 3.80 versus 8.36 s); on ARCENE, preprocessing is about 0.007 versus 9.481 s. MSF preprocessing is lower in every matched core run, with significantly shorter total runtime in three of four core datasets. At population 50 (three guided chromosomes; 6% occupancy), the frozen NSGA-II runs on MADELON, Hill-Valley, and Prostate-GE (seeds 72–81) yield 10/10 paired MSF wins at Gen0 and Gen1. Gen1 mean ΔHV is +0.4165, +0.0312, and +0.4217, respectively (Holm p=0.005859; RBC=1.0 for each).

5.4 Mechanism Attribution and the Prior-Reliability Boundary

Sparse-control separates cardinality placement from Fisher identity. On all five untouched breadth datasets, preserving the mixed-scale cardinalities while randomizing feature identities significantly outperforms Bernoulli Random; restoring Fisher identities adds a further significant gain. Thus both scale placement and relevance identity contribute in this evidence family.

Fisher reliability (mean pairwise Top-K Jaccard at the three injected sizes) was correlated with matched-seed MSF−Sparse-control HV gains in MADELON and ARCENE ($n = 20$ pooled). Spearman $\rho = 0.474$ ($p = 0.035$) for training HV and 0.582 ($p = 0.007$) for held-out HV; neither association was significant within datasets. The dataset-centered held-out $\rho = 0.021$ further cautions against per-run prediction. Top-10 Jaccard was 0.181 on ARCENE versus 0.814 on MADELON. A reliability-triggered fallback to identity-randomized mixed-scale seeding remains future work, not part of frozen MSF.

5.5 Satellite Telemetry as a Structured Feature-Selection Application

The satellite case changes the data domain without changing MSF. Four primary subsystem spaces—Power (267), Thermal (135), Attitude (135), and Data (204)—use the same three-chromosome intervention followed by standard evolutionary search.

Table 4 shows consistent transfer: MSF exceeds matched Random in all 40 paired runs. Mean ΔHV ranges from +0.0380 (Attitude) to +0.1266 (Data), and all four subsystem comparisons remain Holm-significant ($p = 0.007812$). Because wrapper and optimizer are unchanged, the result reflects transfer of the same population intervention rather than a satellite-specific search redesign.

Table 4. Satellite held-out HV (dimensionless; reference (1,1)); higher HV and positive ΔHV favor MSF.

Subsystem	D	MSF HV	Random HV	ΔHV	W/T/L	Holm p
Power	267	0.8054±0.0069	0.7222±0.0120	+0.0832	10/0/0	0.007812
Thermal	135	0.8807±0.0161	0.8343±0.0200	+0.0464	10/0/0	0.007812
Attitude	135	0.9195±0.0105	0.8814±0.0192	+0.0380	10/0/0	0.007812
Data	204	0.9511±0.0019	0.8245±0.0188	+0.1266	10/0/0	0.007812

The frozen 140/582 feature plan reduces median extraction time from 556.11 to 223.58 μs /frame (59.8%) while retaining useful fault evidence under unchanged downstream settings.

5.6 Limitations and Future Scope

Current limits are a heuristic but development-tested scale schedule, a univariate Fisher prior whose identity benefit depends on ranking reliability, validation on two MOEA carriers, and population-50 validation limited to three datasets, one carrier, and the Gen0/Gen1 budget brackets. The satellite evidence is restricted to controlled synthetic telemetry with fault-injection labels; it demonstrates transfer of the unchanged MSF interface and compact-feature utility, but does not establish real-flight or real-orbit accuracy. Future work will target reliability-aware guidance, flight or higher-fidelity data, sparse supervision, and additional learners/carriers.

6 Conclusion

MSF-MOFS treats prior information as a bounded initialization budget: one Fisher ranking supplies three chromosomes at complementary sparsity scales, after which standard MOEA search is unchanged.

Across nine public datasets and two carriers, the frozen construction improves matched-Random search while remaining competitive with richer baselines at lower preprocessing cost. Low-occupancy, Sparse-control, and Fisher-stability diagnostics distinguish scale-placement benefits from reliability-dependent feature identity. In structured telemetry, the same interface also yields a compact feature plan with lower extraction cost. MSF thus provides a reusable initializer for budget-constrained multi-objective feature selection without redesigning the optimizer.

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